

The role of human agency in human-AI co-creativity

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Abstract

Large language models (LLMs) are popular tools for creative ideation, but have been shown to homogenize outputs across users. We test if approaching an AI tool with high human (vs. low) human agency can mitigate this homogenization effect by encouraging people to use AI to augment their creativity, rather than offload it. Participants were experimentally assigned to one of three conditions (high-agency approach with AI access, low-agency approach with AI access, or a human-only control) and generated creative uses for everyday objects. Contrary to our expectations, ideas did not differ in individual-level quality (overall creativity, originality, and usefulness). However, a preregistered similarity-to-centroid analysis and an exploratory cluster analysis provided convergent evidence of AI-induced homogenization among the low-agency condition. Thus, while AI has enabled greater ideational fluency, our research suggests the degree of agency with which people approach their AI tool has downstream consequences on collective creativity.

Keywords: human-AI interaction, creativity, agency, homogenization

The role of human agency in human-AI co-creativity

One of the cornerstones of human cognition and communication is our ability to engage in creative ideation: the process of generating, developing and conveying new ideas. People have long relied on tools to support their creative processes, from physical notebooks to mind-mapping software (Frich et al., 2019). One of the newest and most popular tools to stimulate creativity are large language models (LLMs), which can provide a thought partner to anyone with an internet connection. Fundamentally, LLMs are token predictors that output likely words given a user prompt and context. Their ability to rapidly synthesize massive amounts of text make them useful for a wide range of creativity tasks, and recent reports suggest that people regularly use these systems for tasks related to creativity (Maslej et al., 2025).

The process of people using LLM systems for ideating and communicating creative ideas we refer to as *human-AI co-creativity*. Research to date suggests two important patterns in human-AI co-creativity. The first is that LLM systems perform as well as or even better than humans on commonly accepted creative ideation benchmarks Guzik et al., 2023 and benchmarks for ideation in specific contexts Guo et al., 2025. However, when humans *use* AI systems to be creative, there is a risk of homogenization: People who use AI tend to produce more similar outputs compared to their counterparts who do not—an effect that has been shown in a range of contexts involving open-ended output, such as product design Anderson et al., 2024; Nave et al., 2024 and short-story writing Doshi and Hauser, 2024. In recent years computer scientists have termed this paradox *algorithmic monoculture* Kleinberg and Raghavan, 2021, or, in the case of generative AI, *generative monoculture* Wu et al., 2024. Derived from the notion of monoculture in agriculture, these terms capture the vulnerabilities incurred by mass reliance on a single algorithm. While each AI-assisted individual may generate an idea more creative than what they could achieve on their own, this comes at a risk on the collective level. What is creative with respect to an individual may in fact be trite among a group of individuals using the same AI assistant, especially if

many individuals use AI without bringing their own human experience to the interaction.

In the present study, we examine the extent to which human agency impacts the homogenizing effects of LLMs or preserves the benefits of human-AI co-creativity. Specifically, we explore whether a high-agency approach, in which a person approaches a tool in a way that supports their own agency (rather than replacing it) has downstream consequences on individual- and collective-level ideation outcomes. For example, when people exercise high agency over technologies such as social media, they believe they can control the amount and nature of their use A. Y. Lee and Hancock, 2024. Similarly, encouraging individuals to develop a more intentional orientation towards their AI use may lead them to engage more in creative ideation and increase their likelihood of contributing diverse output to a group.

We test the importance of a high-agency approach in human-AI co-creativity by experimentally manipulating the guidance people receive about AI as a creativity support tool. We compare a group of people who are prohibited from using AI against two treatment groups that were told to use AI: a low-agency condition, in which participants are told to let AI do the bulk of the creative ideation, and a high-agency condition, in which participants are told to leverage their own human creativity in tandem with the AI to develop even better ideas. We assess creativity at both the individual and group levels to explore the effect of agency on both the creativity and diversity of ideas produced in human-AI collaboration, and we examine how various approaches influence the behavioral interactions in the human-AI collaboration.

AI and Creativity

Creativity is complex and difficult to assess, but it can be broadly conceptualized on two levels: individual- and group-level creativity. Individual-level creativity refers to the subjective novelty and value of a single idea considered in isolation, while group-level creativity refers to the diversity of ideas generated by a group of individuals (Doshi & Hauser, 2024). It is possible for individual-level creativity to be high but for group-level

creativity to be low if a vast majority of individuals propose similar ideas.

Creativity at the Individual Level

Benchmarking studies on LLM creativity suggest LLMs can perform well on creative ideation tasks—often better than humans (Guzik et al., 2023). While some sound the alarm for existential threats to human creativity (Hubert et al., 2024), others assert that LLMs still fall short relative to the most creative humans (e.g., (Chakrabarty et al., 2024; Koivisto & Grassini, 2023)). Creativity scholars even suggest that the definition of creativity itself should be updated to account for qualities that LLMs cannot possibly possess, like authenticity or intentionality (Runco, 2025). As with any research involving LLMs, complicating the comparison between humans and models is the rapid pace at which the technology develops and the heterogeneity of model outputs across time, temperature, and system prompts (Haase & Hanel, 2023).

Nevertheless, because of LLMs’ standalone creative potential, people have widely adopted them to generate ideas in schoolwork, product innovation, and even research (Si et al., 2024). To the skepticism of prominent creative thinkers (e.g., (Chiang, 2024)), when the average person uses an LLM to generate open-response content, they tend to produce more creative responses than they would unassisted. For instance, in an experiment in which participants were tasked with writing short stories, participants using ChatGPT produced stories that were evaluated as creative, better written, and more enjoyable than those produced by people who wrote their stories on their own (Doshi & Hauser, 2024). Similarly, participants using ChatGPT to generate creative solutions produced ideas that were rated as more creative than those of participants who used their unaided human intuition or basic web search (B. C. Lee & Chung, 2024). Research that focuses on individual-level dynamics of LLM usage would seem to suggest enhancing effects of ChatGPT for creative tasks.

Homogeneity at the Group Level

While using an LLM may improve the creative performance of each individual relative to their baseline, some analyses of group-level dynamics reveal an important tradeoff. A number of studies have found that the content generated by people working with LLMs tends to be more homogeneous, exhibiting a lower diversity of thought than the set of ideas produced by participants who relied only on their own human cognition (Anderson et al., 2024; Doshi & Hauser, 2024). For instance, although individual ChatGPT-assisted essay performance can be quite high, across the group of students the concepts expressed in the essays are more similar in meaning than the concepts in unassisted essays (Kosmyna et al., 2025).

In creative contexts, the tradeoff between individual-level performance and group-level homogeneity is especially paradoxical: what may seem original to an individual can be banal in a whole group. Studies focusing on group-level effects of LLM usage highlight this paradox. For example, participants who used an LLM for creative writing prompts wrote short stories that were evaluated as more creative, better written, and more enjoyable than those written by their unassisted counterparts. However, the AI-enabled stories were more similar to each other than those written by humans alone (Doshi & Hauser, 2024). Similarly, using AI to brainstorm creative ideas for gifts and household toys reduced the collective diversity of ideas relative to those generated by humans alone (Nave et al., 2024). Although the individual user may perceive the AI’s output as creative and experience real gains relative to their own, unassisted performance, the group as whole suffers from a lack of diversity of thought.

There are a number of technical reasons why LLMs might produce more homogeneous output than a collection of humans. Fundamentally, LLMs are designed to optimize for a response that represents the aggregate of its training data for the particular context invoked by the user (Hicks et al., 2024). This manifests in key model behaviors. For one, LLMs seem to possess an upper limit to the number of new ideas they can generate (Si

et al., 2024). Second, LLMs are designed to predict likely tokens for a given context. When prompted for “creative” ideas, they may provide information related to other instances in their training data in which the notion of creativity has been invoked for the particular use case in the past. Third, as an individual conversation with an LLM proceeds and the context window fills, the LLM is more likely to rehash material it has already produced.

Specific LLM affordances may create an illusion of creative capacity to the people using LLMs for creativity support, making homogenization more likely. A user can input a spare prompt (e.g., “Write an essay on [topic]”) and get a lengthy output from the LLM. Without awareness of this homogenization, however, a society of users relying on the same LLM without variation to their individual behavior may create a cultural problem (Sourati et al., 2025). However, polling data suggests the general public may be unaware of such pitfalls (Poushter et al., 2025). Providing some lay explanations of the mechanisms and consequences of LLM usage may help mitigate passive usage (e.g., copy-pasting without much refinement) and subsequent homogenization at the group level.

Human Agency as a Buffer Against Homogenization

The technical reasons for AI-induced homogenization are quite simple, but mitigating homogenization via human behavior is not. To improve people’s creative work with AI, a growing body of research has focused on improving the technological affordances of AI creativity support tools, including features that allow the user to draw connections between seemingly unrelated concepts (O’Toole & Horvát, 2024) or better steer the model toward their creative vision (Louie et al., 2022). While such solutions seem promising in the laboratory context, human-AI interaction research suggests the need to consider human psychology in addition to design changes to the technology. (Hancock et al., 2020; Tankelevitch et al., 2025).

Another route to disrupting AI-induced homogenization lies in addressing how people understand the role of AI tools in their cognitive processes. People do not use technologies in a vacuum, but rather approach them with implicit beliefs about what the

technology is for and the role it may play in their lives (Crum et al., 2017; A. Y. Lee et al., 2023). In the case of AI, past work suggests that individuals hold distinct beliefs influencing the extent to which they trust AI and rely on its outputs. For instance, people who view AI as a “powerful search engine” are more willing to accept its suggestions than those who see it as a “friend” or a “tool that they use” (Cheng et al., 2026). People also hold specific beliefs about the capabilities of AI to perform specific tasks. Given widespread public narratives about the potential for AI to both turbocharge humans’ creative potential (Shneiderman, 2020) and disempower entire workforces (Dung, 2025), people should hold specific beliefs about the role that AI tools may play in creative thinking processes that influence the outcomes of such collaborations.

A particularly important belief in the context of creative collaboration revolves around the idea of agency: the extent to which individuals perceive themselves, rather than the AI, as the primary driver of creative work. This fundamental psychological need (Bandura, 2001) has been shown to play an important role in other forms of technology use. For example, people with agentic social media mindsets, who perceive social media as a tool they can leverage, use social media to do things that are meaningful, useful, and satisfying to them (A. Y. Lee et al., 2023). In the context of creativity with LLMs, having a high-agency approach would mean perceiving the LLM as a tool to support one’s own human creative tendencies. In contrast, a low-agency approach would mean expecting the LLM to drive most of the creative vision with little human insight.

The Present Study

In the present study, we investigate human-AI co-creativity by examining the role of human agency on individual level creativity and group level homogenization. We experimentally manipulate the type of guidance participants receive before they work with AI on a creativity task. In the experimental conditions, participants are encouraged to either stay in control of the AI tool (high-agency condition) or let it do the heavy lifting (low-agency condition).

To assess creativity, we use a classic creativity paradigm pioneered by psychologists called the Alternative Uses Task (Torrance, 1966), in which a person’s goal is to creatively repurpose everyday objects (e.g., a paperclip). To score well on the task, participants must be able to overcome functional fixedness, or the tendency to focus on the intended uses of the object. Take the brick: It’s easy to think of uses similar to the intended function of a brick, such as “a paperweight” but more creative to propose it as “a chair for a very small child.” Despite a number of limitations, including the focus on idea generation but not evaluation, the task has been widely used because it allows for a relatively standardized way to compare divergent thinking across various conditions. It also relies on little cultural knowledge and writing ability, which is an important task criterion when assessing creativity in a verbal modality.

We test two primary hypotheses. First, given prior literature indicating that LLMs perform well at creativity tasks, we hypothesize that people who are instructed to use AI with a high-agency approach will produce ideas that are rated as more creative at the individual level relative to control and participants with low-agency approaches in terms of an individual’s ideas’ creativity, originality and usefulness.

H1: Ideas produced by participants in the high-agency condition will be rated as more a) creative, b) original and c) useful than ideas produced in the control or low-agency conditions.

The second hypothesis concerned the effect of approach on group-level homogeneity (the similarity of the proposed ideas to each other). In the context of creative ideation, having a high-agency approach should lead people to generate an idea that is not only original in isolation but across a population. We hypothesized that ideas produced by participants in the high-agency condition would be more semantically diverse, or less homogenized, than the ideas produced in the low-agency approach or control conditions.

H2: Ideas produced by participants in the high-agency condition will be rated as less homogeneous than ideas produced in the control or low-agency conditions.

Finally, we also explored how approaches influence the way that participants interact with the LLM system to generate creative ideas. We examine the ways in which high versus low agency approaches influenced how participants prompted the AI system, how often they exchanged turns and words with the system, and finally, if these behavioral changes influenced outcomes at the individual level of creativity and group level of homogenization.

Method

Preregistration

The study was pre-registered on OSF including hypotheses, power analysis, procedure and analytic approach
https://osf.io/2d7fe/overview?view_only=c34f80bd046e4c85aafdee3d2c33a74e.

Participants

We recruited two samples of adult participants from the U.S. to participate in the study: a sample of submitters and a sample of evaluators. Submitters were community college students in the San Francisco Bay Area ($N = 150$; age: $M = 25.1$ years, $SD = 9.2$; 72% female, 24% male, 1.3% non-binary, 2.7% self-described) who participated for research credits. An additional three submitters were excluded for failing an initial attention check. Evaluators were U.S. adults recruited from the CloudResearch Connect platform (Hartman et al., 2023) ($N = 142$; age: $M = 46.7$ years, $SD = 16.3$; 49% female, 51% male) who were compensated financially for their time. Six participants were excluded for failing an initial attention check and five participants were excluded due to idea data logging issues.

Measures and Procedure

Methods were approved by the [ANONYMIZED] Institutional Review Board. All participants provided informed consent.

Submission Phase

Upon consent, participants were asked to read a general description of the Alternative Uses Task (AUT), in which they were asked to generate non-standard uses for three ordinary objects: bubble wrap, frisbee and brick. They were provided with a rubric with our definitions of overall creativity and its two canonical subcomponents: originality and usefulness (Runco & Charles, 1993). After consenting, participants were randomly assigned to one of three conditions that varied whether they completed the idea-generation task with or without access to an AI chatbot (please see Figure A1 for representative excerpts from the instructions shown in each condition). The low- and high-agency conditions provided participants with access to a custom AI chatbot that was designed for the study, which was built on top of gpt-3.5 and embedded into the survey through a web-application. Participants in the low-agency condition were told that “AI recognizes patterns, styles, and connections that would take years of study to master on your own. . . The beauty of AI is that it does the heavy lifting for you.” They were instructed to let AI’s creative power work for them. Conversely, in the high-agency condition, participants were instructed to use AI to expand upon their own ideas and generate ones that are even more creative. This aspect of the manipulation was informed by prior research on growth mindsets, in which interventions encourage strengthening one’s cognitive muscles (Yeager et al., 2019). In the control condition, participants completed the same AUT without access to an AI chatbot. Given that past work suggests that online survey respondents frequently use AI tools to generate answers (Veselovsky et al., 2023), we explicitly instructed participants in the control condition to not use AI in this task. Before the start of the AUT, participants read, “We value your individual human creativity, so please do NOT use AI during the task.” Additionally, copy-paste functionality was disabled using JavaScript code embedded in the Qualtrics survey. Participants were required to spend a minimum of 10 seconds on the instruction page in each condition before proceeding to the AUT.

Occasions

Submitters completed the experiment in one session, with an average completion time of 30 minutes. For each of three objects (frisbee, bubble wrap, and brick), submitters were asked to generate five alternative uses for the object and enter their ideas in form fields. The presentation of the three objects was counterbalanced. Participants in the high-agency and low-agency conditions had access to ChatGPT through an embedded web-app in the Qualtrics survey; they entered their alternative uses below the web-app interface. Participants in the control condition had no web-app, just the fields to enter their alternative uses for each object. In line with prior research on induction with the AUT Madore et al., 2016, all participants were given as much time as they needed to complete the task. This avoided penalizing the participants in the low-agency and high-agency conditions for taking time to work with the AI chatbot. Across conditions, submitters produced a total of $n = 2,250$ ideas.

After completing the AUT, submitters were asked to rate their own ideas in terms of overall creativity and its subcomponents: originality, and usefulness (Boden, 1998). Each submitter was provided with the following rubric to guide their scoring (See Table A1). To reduce survey fatigue, each participant was asked to rate one of the ideas they had submitted for a brick, one for bubble wrap, and one for frisbee. These ideas were randomly selected from their set of submissions. (e.g., “One of your alternative use ideas for a frisbee was a [dish]. Provide your ratings of this idea below.”) Responses to the creativity, originality, and value questions were on a Likert scale from 1 = Not at all to 5 = Maximally. To gauge participants’ perceptions of the uniqueness of their ideas, we also asked, “Do you think many other people came up with this idea as well?” Response options included “No one thought of this”, “Some people thought of this”, “Most people thought of this”, and “Everyone thought of this.”

Finally, all submitters provided a brief description of their idea generation process in a text box. Participants in the high-agency and low-agency conditions were asked to

self-report the extent to which the AI helped them with the creative process (1 = Strongly Disagree to 7 = Strongly Agree; B. C. Lee and Chung, 2024, the extent to which they perceived AI as a conversational partner (1 = Strongly Disagree to 7 = Strongly Agree; $\omega = 0.86$) B. C. Lee and Chung, 2024, and their intrinsic motivation during the AUT (how interesting, enjoyable, fun, and engaging it was; 1 = Not at all to 7 = Completely; $\omega = 0.92$) Shin and Grant, 2019. All participants answered demographic questions before being debriefed on the purpose of the study and granted research credits for their participation.

Evaluation Phase

Next, we asked our evaluators to rate the compliant ideas from the submitters. Evaluators were shown the same definitions of creativity, originality, and usefulness as the submitters, which served as their rubric upon which to score the ideas. Each evaluator rated a random subset of 15 of the total set of ideas. The submitted ideas were stratified by object type, such that each evaluator rated five ideas for bubble wrap, five ideas for a brick, and five ideas for a frisbee. For each of the 15 ideas they saw, the evaluators rated it in terms of overall creativity, originality, and usefulness, resulting in 45 ratings produced by each evaluator. Due to our randomization procedure, some ideas were evaluated multiple times, while others were never evaluated and thus were excluded from this analysis. All participants answered demographic questions before being debriefed on the purpose of the study and compensated for their participation.

Manipulation Checks

We conducted manipulation checks to ensure that submitters followed the instructions for their conditions. Using the chat log data, we identified trials in which participants in the high-agency and low-agency treatment conditions did not use the AI tool as instructed. In 24% of low-agency trials and 38% of high-agency trials participants did not use the AI tool. Object trials from the high- and low-agency conditions were excluded if they did not include chat data. On the participant level this resulted in 12 participants in the high-agency condition and 10 participants in the low-agency condition

being completely excluded from the analysis.

Data Analysis

Data Structure

Each participant produced five ideas for three objects (brick, bubble wrap, frisbee). For participants in the high-agency and low-agency conditions, the survey was designed such that each object corresponded to one human-AI interaction, with chat data logged using JavaScript.

Analytic Approach

We assessed the creative qualities of participants' submitted ideas at both the individual level and the condition level. To assess creativity, we examined the ratings from evaluators at the individual participant level. To assess homogeneity, we measured the semantic similarity of ideas generated by participants at the condition level (cf. Anderson et al., 2024). We used the sentence embedding model all-MiniLM-L6-v2 (Reimers & Gurevych, 2019) to generate embeddings in a 384-dimension embedding space. Note that cosine similarity can take on values from -1 (semantically opposite) to 1 (semantically identical), although in practice the lower bound is closer to 0.

Statistical Approach

We note two key deviations from our preregistration. Although the preregistered target number of idea submitters was $N = 450$ (to attain 95% power with a small effect size), external constraints on recruitment duration resulted in a final sample of $N = 150$. The results of our simulated a priori power analysis indicated that we would be powered to detect a medium effect ($\eta_p^2 = .06$) at 80% power. Second, upon collection of the data, analysis of participants' chat-log data indicated that a substantial portion of the participants in the high-agency ($N = 12$) and low-agency ($N = 10$) conditions did not comply with the task instructions and were therefore excluded from the final analyses. Given these limitations, our sample reported below is smaller than our preregistered target

sample.

Using the lme4 package in R (Bates et al., 2015), for each of the preregistered outcome metrics we fit linear mixed effects models with condition as the fixed effect and participants and objects as random effects.

Results

Individual-Level Creativity

To examine the creativity of the submissions, we conducted separate mixed-effects linear regression models to predict each of the main dependent variables (creativity, originality and usefulness ratings from third-party evaluators) from the fixed effect of condition, with random effects for submitters and objects (bubble wrap, brick, frisbee). Contrary to our first hypothesis (H1), there were no significant differences in overall idea creativity or originality across the treatment conditions relative to control, $ps > .60$, as shown in Figure 1. Regression results for all three metrics of individual-level outcomes are shown in Table A2. Ideas from participants in the low-agency condition were rated as significantly more useful, $\beta = 0.20$, $p = .04$, which we address in the Discussion.

Homogeneity

Our second hypothesis (H2) predicted that participants in the high-agency condition would have lower levels of homogenization across their submitted ideas than participants in the low-agency condition. We operationalized group-level homogeneity using the most common approach from prior research: the cosine similarity of each idea to the centroid embedding of the other ideas in the same condition for the same object (Doshi & Hauser, 2024). Specifically, for each participant we randomly sampled one idea they had generated for each object (brick, bubble wrap, frisbee) and computed $1 - \text{cosine}(\text{ideaembedding}, \text{centroidembedding})$. The centroid embedding used a leave-one-out approach: the average embedding of all ideas other than the one being compared.

A linear mixed-effects model with condition as fixed effect and participants and

objects as random effects revealed that, consistent with H2, there was a significant effect of experimental condition, $F(2, 121.67) = 3.15, p < .05, \eta_p^2 = .07$ (see Table A3). Pairwise comparisons with Tukey’s HSD revealed that participants in the high-agency condition submitted ideas that were less similar to the condition-object centroid than participants in the low-agency condition, $t(125) = -2.45, p < .05$, indicated by greater cosine similarity scores in the low-agency condition. In contrast, there were no differences in idea homogeneity between the high-agency condition and the control condition, $t(122) = 1.87, p = .15$. This pattern of results suggests that while participants in the low-agency and control conditions produced ideas that were similarly homogeneous relative to the "average" idea in their respective conditions, participants in the high-agency condition were buffered against homogeneity via their instructions.

Although the centroid approach is standard among the literature on AI-induced homogenization in creative domains, it relies on the assumption that the structure of ideas can be distilled into a single point estimate of an average. However, as this approach aggregates across embeddings, it fails to capture the inherent variability in the structure of each condition’s collective ideas. Furthermore, sentence embeddings are themselves a product of the sentence embedding model’s weights, but treating them as standalone component scores assumes the particular values are reliable enough to derive meaningful difference scores (i.e., cosine distance). To address this concern, we conducted an exploratory clustering analysis (HDBSCAN; McInnes et al., 2017) in which sentence embeddings allowed us to identify higher-level categories of ideas, ideas that did not clearly fall into a single category of ideas, and evidence of participants glomming onto a single category—our key indicator of homogeneity at the group level.

We used three metrics to quantify the results of the clustering analysis: (1) number of unique clusters, (2) noise ratio, and (3) coefficient of variation. The number of unique clusters served as a proxy for the number of unique categories of uses proposed by participants in each condition. Noise ratio is the proportion of ideas that did not fall under

any cluster, serving as a proxy for the "outlier" ideas indicative of greater diversity. The coefficient of variation (CV; standard deviation of cluster sizes divided by the mean cluster size) indicates the extent to which the clusters are equal in size; a higher CV suggests greater imbalance in cluster sizes. Given the size of our dataset, we set the minimum cluster size to 6, minimum samples to 1, and cluster selection epsilon to 0.25. As shown in Figure 3, the three conditions showed distinct patterns in these metrics.

To test whether the observed differences in clustering patterns were statistically significant, we conducted permutation tests ($n = 5,000$ permutations) comparing the maximum and minimum condition means for each metric. For the number of unique clusters, the observed difference between conditions was 3.33, a difference driven by the statistically higher number of clusters in the control condition relative to the low-agency condition, $p = .04$. For noise ratio, the observed difference was 0.22, again driven by a higher noise ratio in the control condition relative to the low-agency condition, $p = .02$. For coefficient of variation, the observed difference was 0.75, driven by the low-agency condition having a significantly higher coefficient of variation than the control condition, $p = .005$. Taken together, the clustering results suggest that our experimental manipulation resulted in homogeneity outcomes significantly different from what would be expected by chance.

Discussion

In the present study, we investigated how psychological approaches to AI usage affect creativity at both the individual and group levels. By experimentally assigning people to read one of three common forms of guidance about AI usage (stay in control of the AI tool, let the AI tool do the heavy lifting, or don't use AI at all), we elucidated the effect of human agency. First, at the individual level we observed no significant differences in the ideas as assessed by third-party evaluators' subjective ratings of overall creativity, originality, and usefulness. However, in line with prior research on the "paradox" of human-AI co-creativity our group-level analysis of the semantic similarity of ideas by condition revealed that low-agency participants were significantly more homogeneous than

their high-agency counterparts. Our exploratory cluster analysis corroborated this finding: Specifically, the low-agency condition had a significantly greater coefficient of variation than the other high-agency and control conditions, reflecting both a low number of unique clusters and a low noise ratio, or the proportion of ideas that might be considered unique. Together, our preregistered and exploratory homogeneity analyses provide converging evidence for the hypothesis that using AI with low human agency induces greater homogeneity among the group.

Our findings are consistent with previous research on human-AI co-creativity, whereby people who work with AI tend to produce more similar creative output than people who rely on their own minds (Luchini et al., 2025). On a broader level, the present study contributes to a growing body of research suggesting that LLMs flatten human variability (Wang et al., 2025), which has important implications for creativity. Critically, participants in the high-agency condition were no more homogeneous than the control group, suggesting that approaching an AI tool with one’s human background improves group-level creative outcomes.

We also contribute to creativity research more broadly through our analytic approach and classification of people’s collaborative strategies. In our paradigm we captured not only the creative product (participants’ ideas), but the process by which participants worked with AI to generate those ideas (Green et al., 2024). By exploring the prompting strategies people used to solicit creative help from AI, we uncovered strategies similar to those in research on stimulating creativity among humans. For example, the Exploratory category of prompts (in which participants specify a context in which the object may be used) aligns with the notion of “creative constraints” (Medeiros et al., 2025), whereby people actually generate more creative ideas when they have a degree of constraint on the type of output they are allowed to create. The high-agency condition showed more self-imposed creative constraints, which may be a distinct pathway to enhanced creativity. In such an open-ended task as the Alternative Uses Task, the

Exploratory prompting strategy—used at a higher rate by our high-agency participants compared to our low-agency participants—may funnel a greater number of unique ideas into the set of generated ideas.

Amid widespread adoption of AI, human-centered AI research consistently calls for “designing AI systems that support human self-efficacy, promote creativity, clarify responsibility, and facilitate social participation” (Shneiderman, 2020). Additional efforts are needed to combine both design-level interventions designed to support agency in human-AI interactions with individual-level interventions that can help everyday people approach and leverage AI tools without deferring their own autonomy. For instance, psychological interventions that remind individuals that they should retain ownership of their work and that there are long-term benefits to exercising their cognitive abilities may help establish a high-agency foundation for people using AI during developmentally critical periods (i.e., early adolescence, upon first adoption) (Walton & Wilson, 2018). Furthermore, AI literacy interventions that explicitly increase individuals’ knowledge about the inner workings, capabilities, and limitations of modern AI tools may similarly help people feel a greater sense of control over their interactions (Long & Magerko, 2020).

Limitations

The results of this study should be interpreted with respect to its limitations. While our study suggests that human agency plays a role in how people engage in creative ideation with AI, the generalizability of our findings is limited in several key ways. First, we focused on participants’ ability to engage in divergent thinking, which is but one of many ways of conceptualizing and assessing creative thinking (Byrge et al., 2025). Furthermore, the Alternative Uses Task is just one way to assess this specific type of “little-c” or “everyday creativity” (Ivcevic & Grandinetti, 2024), so it is unclear whether human agency exerts a similarly strong causal influence on other creative tasks, such as artistic work, or other types of creative thinking (e.g., convergent thinking). Furthermore, we rely on third-party evaluators to subjectively assess aspects of creativity. Although we

provided each evaluator with a rubric upon which to score ideas, evaluators likely varied in their criticality, potentially introducing measurement error. Additionally, as we focused our study on the human side of the human-AI interaction, computational work is needed to probe model behavior for the Alternative Uses Task across various parameters, such as temperature, which is known to affect the randomness of generative models (Peeperkorn et al., 2024).

Notably, a high rate of noncompliance was observed in the AI-access conditions. A closer examination of participant feedback indicated that this noncompliance may have been linked to defiance (e.g., the participant refused to use AI on principle). For example, reading about AI’s potential to control the user may have motivated some participants to refuse its help altogether—an unintended consequence of our experimental manipulation. Such psychological reactance would be in line with prior research on the forced adoption of self-service technology (Feng et al., 2019) or the broader phenomenon of aversion to AI in conventionally human domains (Candrian & Scherer, 2023). While this somewhat compromised the random assignment, it also revealed real reactions to AI and digital literacy guidance that may occur in the wild, especially for a context as central to human identity as creation (Morewedge, 2022). Given that the intrinsic motivation to complete the AUT was fairly high across conditions, participants may have felt that the AI chatbot interfered with their ability to enjoy the experience of generating ideas on their own.

Future Directions

Our results support several promising directions for future research. It is an open question whether the idea homogenization we observed would result in the kind of vulnerability to system shocks as described by Kleinberg and Raghavan, 2021; future work could document the implications of such homogenization in a higher-stakes domain than everyday idea generation. Furthermore, we captured creative ideation at a single timepoint, but it will be important to track how engagement with ChatGPT may change over time. As highlighted by several researchers investigating effects of ChatGPT usage on

cognition more broadly, it is unclear whether ChatGPT can be conceptualized as an “addictive” medium that impairs critical thinking and other mental processes (Yankouskaya et al., 2025) or a tool to enhance it (Suriano et al., 2025). A longitudinal study could shed light on changes in creative performance over time as a function of psychological approaches to AI. Additionally, future studies might strengthen the manipulation of ChatGPT usage by having participants in the high-agency and low-agency conditions behave differently, not just think differently (O’Toole & Horvát, 2024). For example, instead of simply providing a chatbot for participants to use, future studies might embed a tutorial in which participants have the opportunity to ask questions to ChatGPT and push back on its suggestions for creative ideas. Finally, it may be fruitful to explore the natural tendencies of the individuals whose ideas were rated as the most creative, original, and useful. Individuals vary in their baseline levels of creativity and it will be beneficial to understand the boundary conditions of a similar intervention on downstream consequences like diversity of thought. In practice, however, selection effects may be at play. Indeed, some of the most creative individuals may not use AI at all for their creative work, either because they are better at the creative task on their own or because they refuse to use it on principle, like some of our participants.

Conclusion

AI systems have lowered the barrier to entry for many forms of creative work and expression. While this has democratized creative ideation for people who may feel the need for creativity support, our study suggests that without an agentic approach people who use large language models may unwittingly contribute to an algorithmic monoculture. As illustrated by even our modest intervention, mass reliance on the same model may have a more insidious effect on the everyday users who count on an idea being just a prompt away.

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Contributions

All authors conceptualized the study. S.H.W., Y.Y., A.Y.L., and J.H. selected measures. S.H.W. collected and analyzed the data. Y.Y. implemented the chatbot in the survey and analyzed chat-log data. S.H.W., A.Y.L., and J.H. wrote the paper. J.H. and K.N. supervised the project.

Competing Interests

Authors with BetterUp affiliations are paid employees of BetterUp Inc.; A. Lee and J. Hancock are paid consultants within BetterUp Labs at BetterUp, Inc. S.H.W. and Y.Y. do not have any affiliation with BetterUp and has no conflicts of interests to declare.

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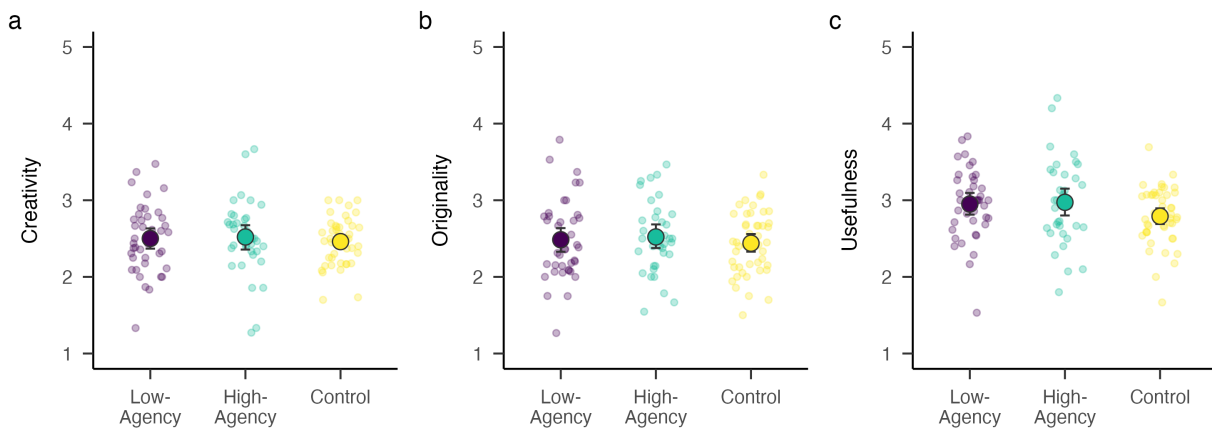
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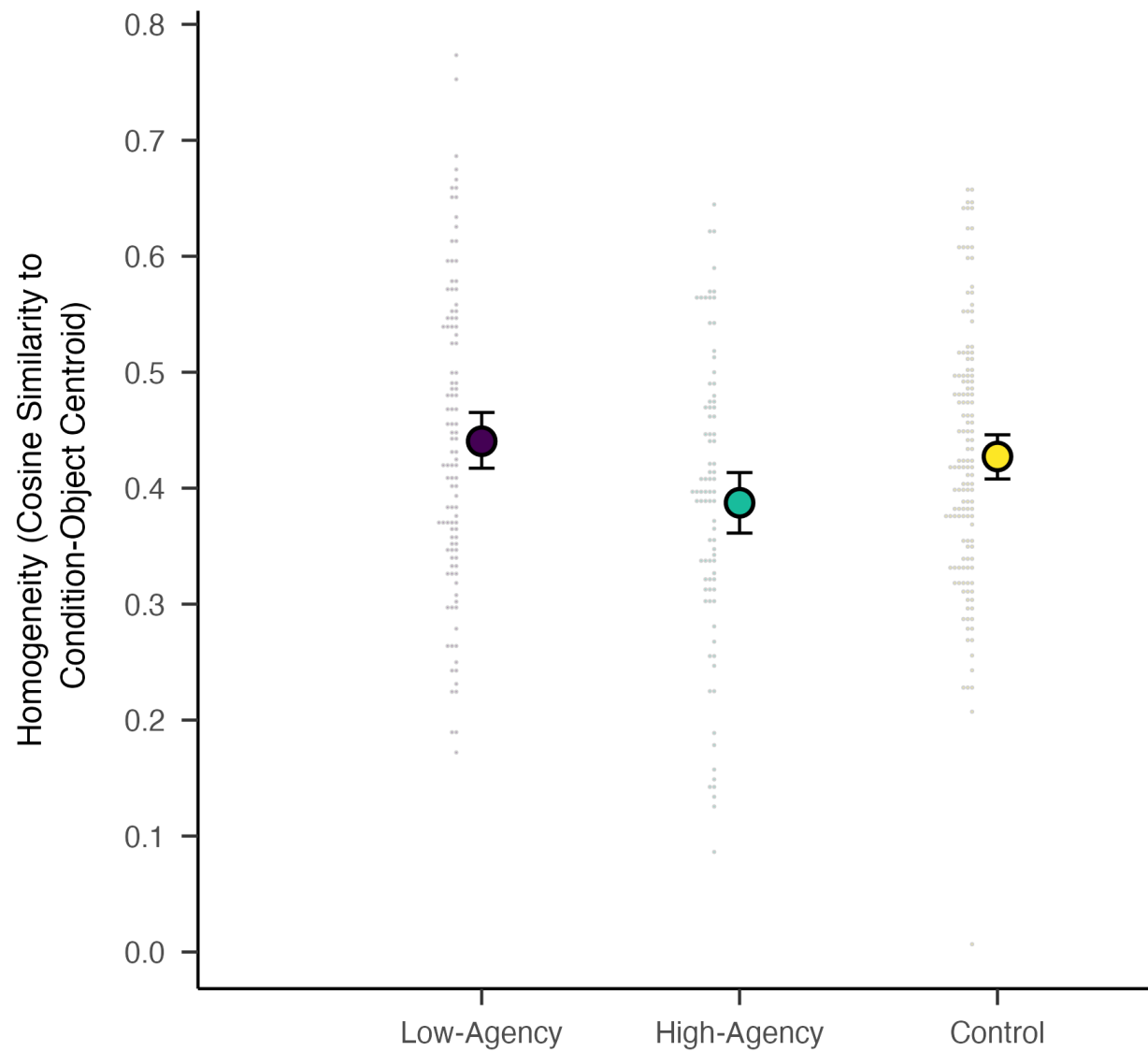
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Figure 1*Individual-Level Creative Outcomes Across Conditions*

Note: Large colored dots indicate condition means, with bootstrapped 95% confidence intervals. Jittered points in the background indicate the distribution of cosine similarity scores between each idea embedding and its condition-object centroid embedding.

Figure 2*Homogeneity (Similarity to Condition-Object Centroid)*

Note: Colored dots indicate condition means, with bootstrapped 95% confidence intervals. Light gray points indicate the distribution of cosine similarity scores between each idea embedding and its condition-object centroid embedding.

Figure 3*Clusters Across Conditions and Objects*

Note: Clusters were identified for each combination of object and condition using HDBSCAN and visualized in 2D using t-SNE. Ideas that were classified as “noise” by the clustering algorithm appear as light gray points, while ideas from unique clusters each appear as a brightly colored point.

Appendix A

Experimental Platform and Paradigm

Web App and Language Model

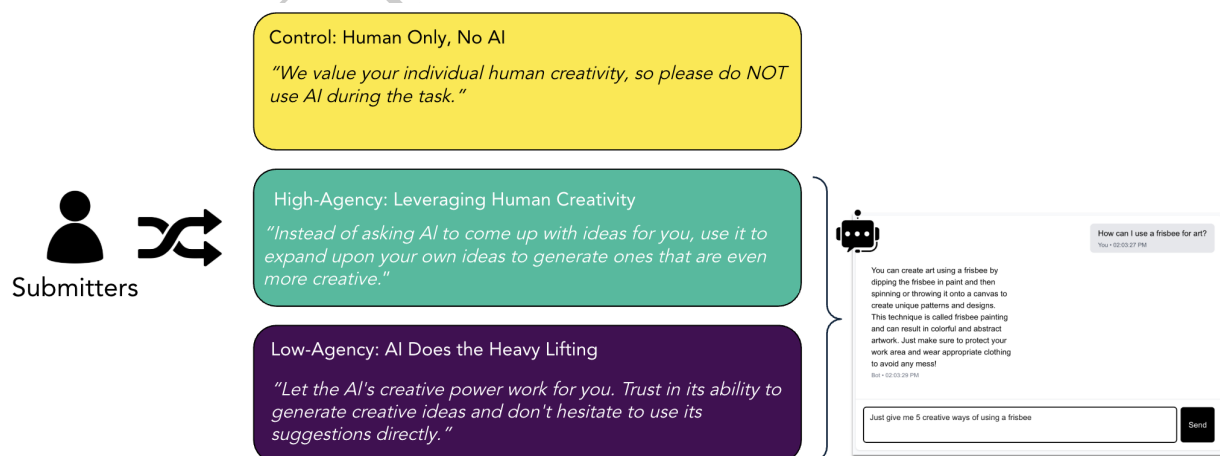
We developed a simple web application in Vercel which emulated the generic AI chatbot experience (e.g., ChatGPT). The interface was embedded in the Qualtrics survey using Javascript. Participants' chat interactions with the AI, including message sender (i.e., AI or user), message content, and timestamp, were logged as responses together with other survey data through the Qualtrics API. We used the gpt-3.5 model, with the temperature parameter set to 0.7, in line with other research on ChatGPT for creative tasks (e.g., Ashkinaze et al., 2024).

System Prompt. The model was given the following system prompt for all interactions with participants: "You are a helpful AI assistant in a chat interface. Be conversational, friendly, and helpful. Keep responses concise but informative."

Experimental Paradigm

Figure A1

Experimental Paradigm



Creativity Scoring Rubric

Table A1

Creativity Scoring Rubric, Provided to Both Submitters and Evaluators

Metric	Definition Provided to Participants
Overall creativity	This is your single, overall impression. Does the idea strike you as clever, ingenious, or highly innovative? Does it provide a surprisingly good solution to a real-world challenge?
Originality	Originality means an idea is new, uncommon, or unique. It’s something that hasn’t been done before in the exact same way. Think of it as the “surprise” factor. To guide your originality scoring, you can ask yourself: “Is this the first thing that comes to mind, or is it something I wouldn’t have thought of?”
Usefulness	Usefulness means the idea is effective, appropriate, or valuable in some way. It has to work, make sense, or solve a problem. In short, it must have a purpose. An idea that is impossible or nonsensical is not useful.

Table A2*Linear Mixed Models Predicting Creativity, Originality, and Usefulness*

	Creativity	Originality	Usefulness
(Intercept)	2.537*** (0.103)	2.514*** (0.136)	2.955*** (0.087)
High-Agency	−0.040 (0.091)	−0.010 (0.110)	−0.026 (0.101)
Control	−0.082 (0.081)	−0.077 (0.098)	−0.162 (0.090)
SD (Submitter intercept)	0.225	0.328	0.264
SD (Object intercept)	0.143	0.199	0.095
SD (Residual)	1.134	1.186	1.223
Num.Obs.	1709	1709	1709
R2 Marg.	0.001	0.001	0.004
R2 Cond.	0.053	0.096	0.053
AIC	5358.4	5543.6	5622.3
BIC	5391.1	5576.3	5655.0
ICC	0.1	0.1	0.1
RMSE	1.12	1.16	1.21

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The Low-Agency condition is set as the reference level.

Table A3*Linear Mixed Model Predicting Similarity to Condition-Object Centroid*

	Cosine Similarity to Centroid
(Intercept)	0.441*** (0.015)
High-Agency	−0.050* (0.021)
Control	−0.014 (0.019)
SD (Submitter intercept)	0.057
SD (Object intercept)	0.010
SD (Residual)	0.109
Num.Obs.	337
R2 Marg.	0.025
R2 Cond.	0.239
AIC	−441.0
BIC	−418.1
ICC	0.2
RMSE	0.10

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The Low-Agency condition is set as the reference level.

Participant–AI Semantic Similarity

To quantify the similarity between participant-generated ideas and AI-generated suggestions, we first used gpt-5-mini to extract AI’s suggested uses from the chat log data (e.g., extract the word “doorstop” from “use a brick as a doorstop.”). Second, for each participant, we compared each of their five submitted alternative uses to the list of AI ideas from that participant’s conversations with AI. We used a sentence embedding model (all-MiniLM-L6-v2) to calculate cosine similarity scores between the participant’s ideas and the AI’s suggestions, and used the maximum similarity value as a proxy for the closest match between a participant’s idea and an AI suggestion. We averaged these maximum similarity scores across all ideas for each participant, yielding a single participant–AI idea similarity score for each participant. The average participant–AI similarity score was not significantly different across the high-agency and low-agency conditions, $ps > .05$.

Appendix B

Additional Results

Behavioral Outcomes

To probe the nature of participants' prompts in finer detail, we conducted several exploratory analyses on how the agency mindsets influenced the human-AI collaboration. First, to examine if the mindset manipulation led participants to prompt the AI chatbot differently, we conducted an exploratory qualitative analysis of participants' initial prompts, specifically the first message sent to the chatbot in each conversation, and developed a coding scheme to categorize these initial prompts. We first used an inductive approach to manually code 20% of the initial prompts. Then, we identified a relevant framework from the creativity literature to guide the rest of the coding scheme and prompted gpt-5-mini to classify the prompts according to its conceptual definitions and examples. Two prominent themes emerged: Direct Asks, which referred to cases where the participant prompted the AI system with the task instructions, and Exploratory, which referred to cases where participants asked the AI system to develop ideas based on a more specific category (see Section B for full definitions, examples, and descriptive data). As shown in Figure B1, participants in the low-agency condition used Direct Asks at a higher rate than participants in the high-agency condition, while they used Exploratory prompts at a lower rate than participants in the high-agency condition.

Finally, we conducted a series of independent-samples t-tests comparing the following features of the participants' conversations with AI: number of conversational turns, chat duration (in seconds), average word count per turn, and number of copy-paste events. These values did not significantly differ between the high-agency and low-agency conditions, $ps > .17$, suggesting that while the mindset manipulation affected the types of initial prompts participants made, they did not manifest in the behavioral indicators from the logged chat data.

Treatment Noncompliance

Analysis of participants’ behavioral interactions with the AI chatbot indicated noncompliance in both the high-agency and low-agency conditions. In 24% of the low-agency trials and 38% of the high-agency trials, participants did not use the AI web-app. Examination of qualitative participant feedback indicated that in some cases, this noncompliance may have been due to a lack of clarity in the task instructions, but in most cases this was due to the participant’s deliberate choice to refuse help from AI on principle (e.g., a participant in the high-agency condition who explained that they did not use the chatbot because “AI takes away my critical thinking and would make this boring”).

Participants’ Prompts to the AI System

Prompt Coding Procedure

We developed a coding scheme to categorize participants’ initial prompts to the AI system. First, we randomly selected 20% of the participants and coded their prompts based on patterns in the subset. A vast majority of initial prompts simply repeated the task instructions (e.g., “Give me creative uses for a brick.”). The first and second author coded these as “Direct Asks.” For the remaining initial prompts, we used inductive coding and reached $\kappa = 0.89$. Disagreements were resolved through discussion.

To guide our subsequent coding, we drew upon Boden’s (1990) framework of three types of creativity: Exploratory, Combinational, and Transformational. Exploratory creativity discovers novel ideas within a constrained space; combinational creativity merges existing concepts to form a new one; transformational creativity completely alters the rules of the conceptual space.

We used GPT-5-mini to classify participants’ first prompts in each conversation with the AI.

System Prompt to GPT-5-mini

CATEGORY_DEFINITIONS

You are analyzing user prompts sent to an AI assistant about alternative uses for objects.

Classify each prompt into exactly one of the following categories:

1. **DIRECT__ASK:** User directly asks the AI to generate ideas.

Examples:

- “5 creative ways to use a brick”
- “Please come up with 5 creative ways to use a frisbee”
- “give me alternative uses for bubble wrap”

2. **TRANSFORMATIONAL:** The restructuring or alteration of the fundamental rules and constraints of a conceptual space. User imagines changing the object’s fundamental properties or nature.

Examples:

- “A brick that uses brain signals/waves to mold onto something we are imagining, like durable play dough”
- “if I melted bubble wrap what would it become”
- “what if a brick could change shape”

3. **EXPLORATORY:** Searching a defined space. This includes information seeking or exploring using an object under a specific context.

Examples:

- “Bubble wrap is the same as pop its?”
- “come up with 5 creative ways to use a frisbee, that have to do with pets or food”
- “what properties does a brick have”
- “how does bubble wrap work”
- “can bubble wrap be used as a pillow”

4. **COMBINATIONAL:** Generating new ideas by combining or connecting existing, familiar concepts. Putting together known ideas in novel ways.

Examples:

- “how could I use a frisbee to replace or aid a piece of machinery”
- “can I use a brick with cement”
- “combine bubble wrap with tape”
- “could I grind up a brick and mix it with some things to make clay or cement”

Important: Each prompt must fall into exactly one category. Choose the category that best fits the primary intent of the prompt.

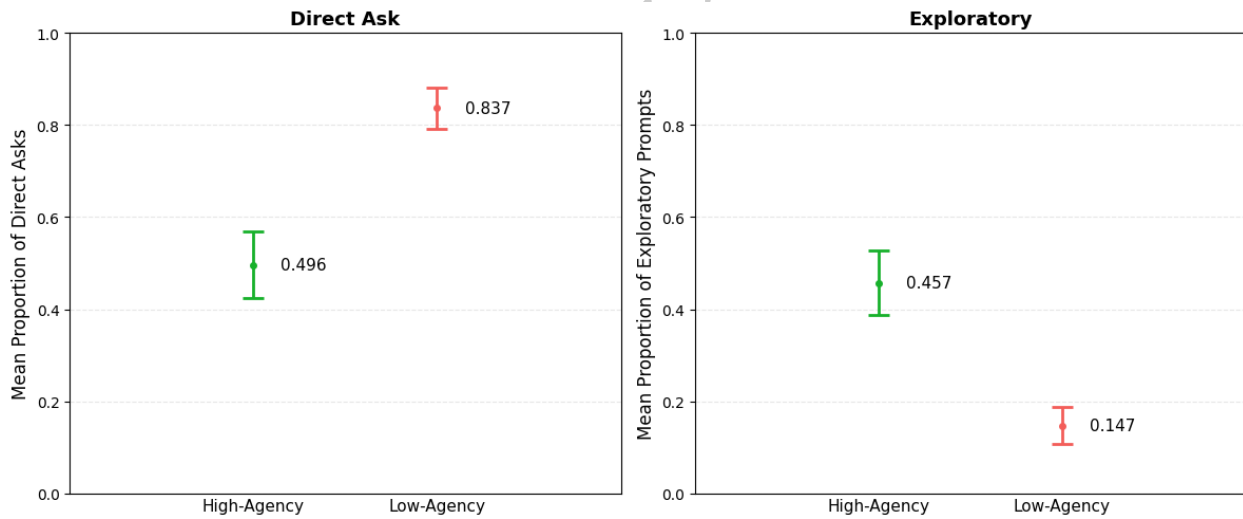
Respond with only one of the following labels:

DIRECT_ASK EXPLORATORY TRANSFORMATIONAL COMBINATIONAL

For each participant, we calculated the proportion of their initial prompts (across up to three objects) falling into each category. We then compared the rate of each type of prompt between participants in the High-Agency and Low-Agency conditions.

Figure B1

Prompting Across AI-Access Conditions



Note: Transformational and Combinational prompts were uncommon in the sample (< 2% in each condition), and thus are omitted from the figure.

Prompt Classification Results

Direct Ask. Participants in the Low-Agency condition used significantly more Direct Ask prompts ($M = 0.84$, $SD = 0.29$) than participants in the High-Agency group ($M = 0.50$, $SD = 0.45$), $t(79) = 4.01$, $p < .001$, $d = 0.91$.

Exploratory. Participants in the High-Agency condition used significantly more Exploratory prompts ($M = 0.46$, $SD = 0.44$) than participants in the Low-Agency condition ($M = 0.15$, $SD = 0.26$), $t(79) = 3.81$, $p < .001$, $d = 0.86$.

Transformational and Combinational. Transformational and Combinational prompts were rare in the sample. There was no significant difference in the rate of Transformational prompts between the Low-Agency condition ($M = 0.01$, $SD = 0.05$) and the High-Agency condition ($M = 0.02$, $SD = 0.11$), $t(79) = -0.50$, $p = 0.620$, $d = -0.11$. There was no significant difference in Combinational prompts between the Low-Agency condition ($M = 0.01$, $SD = 0.05$) and the High-Agency condition ($M = 0.03$, $SD = 0.11$), $t(79) = -1.19$, $p = 0.24$, $d = -0.26$.